

Pavel Zakharov

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Education

TU Dortmund, Germany, Department of Mathematics, PhD Program	2023 – present
Higher School of Economics, Moscow, Russia, Data Science, Math in Machine Learning, Master Program	2021 – 2023
Higher School of Economics, Moscow, Russia, Faculty of Computer Science, Applied Mathematics and Informatics, Bachelor Program	2017 – 2021
Physical-Mathematical School 2007, Moscow, Russia, Graduated with Golden Medal.	2017

Positions

TU Dortmund, Department of Computer Science, Research Assistant	2023 – present
University of New South Wales, Research Internship	2025
HSE University, Laboratory of Theoretical Computer Science, Research Assistant	2019 – 2023
Yandex LLC, Yandex.Routing, Yandex.Adtech, Software engineer	2019 – 2020

Research

Preprints

Fluctuations of the Ising free energy on Erdős-Rényi graphs, A. Coja-Oghlan, D. Kaaser, M. Rolvien, P. Zakharov, K. Zampetakis.	2026
The number of random 2-SAT solutions is asymptotically log-normal A. Chatterjee, A. Coja-Oghlan, N. Müller, C. Riddlesden, M. Rolvien, P. Zakharov, H. Zhu, To appear in <i>Theory of Computing</i> .	2024

Journal publications

A note on the maximum clique LP relaxation, P. Zakharov, <i>Operations Research Letters</i> 66 (2026) 107425.	
The random k-SAT Gibbs uniqueness threshold revisited, A. Chatterjee, A. Coja-Oghlan, C. Greenhill, V. Pfenninger, M. Rolvien, P. Zakharov, K. Zampetakis, <i>Combinatorics, Probability and Computing</i> (2026) 1–53.	
WalkSAT is linear on random 2-SAT, P. Berenbrink, A. Coja-Oghlan, C. Cooper, T. Götze, L. Hintze, P. Zakharov, <i>SIAM Journal on Discrete Mathematics</i> 39 (4) (2025) 1939–1952.	
Fractional colourings of random hypergraphs, D. Shabanov, P. Zakharov, <i>Russian Mathematical Surveys</i> 78 (6) (2023) 183–184.	
Asymptotic normality of the giant component size in a random bipartite graph, D. Shabanov, P. Zakharov,	

Trudy MIPT **15** (2023) 23–32 (in Russian).

On the maximal cut in a random hypergraph,

D. Shabanov, P. Zakharov,

Doklady Mathematics **104** (2021) 336–339.

Talks

31st British Combinatorial Conference, 2026
Cardiff, United Kingdom.

Randomness in Computing, 2026
Sheffield, United Kingdom.

**Athens-Dortmund-Hamburg Workshop on Graphs, Networks, Optimization,
and Statistical Inference,** 2026
Corfu, Greece.

Midwinter meeting in Discrete Probability, 2026
Umeå, Sweden.

Algorithmic Decision Theory Reading Group, 2025
Sydney, Australia.

22nd International Conference on Random Structures and Algorithms, 2025
Vienna, Austria.

28th International Conference on Randomization and Computation, 2024
London, United Kingdom.

Algorithms, Dynamics, and Information Flow in Networks Seminar, 2024
Online.

European Conference on Combinatorics, Graph Theory and Applications, 2021
Online.

Research grants

Co-investigator on “Extremal properties of random graphs”. Russian Foundation for
Basic Research and Iran National Science Foundation, no. 20-51-56017 2020 – 2022

Co-investigator on “Structural properties of random graphs and hypergraphs”.
Russian Science Foundation, no. 22-21-00411 2021 – 2023

Teaching

TU Dortmund, Department of Computer Science 2026
Complexity Theory — Design of study materials, tutoring

Higher School of Economics, Faculty of Computer Science 2021 – 2024
Theoretical Computer Science — Lecturer, course design

Mathematical Statistics — Design of study materials, tutoring

Algorithms and Data Structures — Design of study materials, tutoring

Theory of Computation — Tutoring

Convex Programming and Approximation Algorithms — Tutoring

Probability Theory — Tutoring

Sberbank 2021/22
Graph Theory – Tutoring

Service and outreach

**ADYN Summerschool on Algorithms, Dynamics, and Information Flow in
Networks** 2026
Organizing

“Combi meets Stat Physics” reading group, UNSW 2025
Organizing

Peer reviewing and subreviewing

Annals of Applied Probability, Australasian Journal of Combinatorics, Combinatorica, Random Structures and Algorithms, STACS	
Project programme in mathematics & theoretical computer science, “Sirius” Educational Center	2022
Mentor	
Mathematics summer school “Kostroma open”	2017
Invited teacher	

Other

Ilya Segalovich scholarship	2021-2022
Awarded by Yandex, €10K	